ZS-Q9
Complex Time-Delay Locking on the i-Tetration Fixed Point: A Two-Coordinate Spectral Reconstruction Bridge for Sub-Unitary Scattering Systems

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Verification: 52/55 PASS (3 PARTIAL on Q9.5 synthetic, registered) | Zero Free Parameters | Six structural results + Corollary Q9.4a | Two-Coordinate Reconstruction PROVEN at 50-digit | Lemmas Q9.3a/b correct v1.1 sign error | Anti-Numerology $\ln \Lambda = 8.94$ (decisive evidence)

§0. Abstract

We present ZS-Q9 v1.2, a mathematically strengthened framework that lifts the i-tetration fixed point $z^* = 0.4382829... + 0.3605925...i$, previously locked at five self-consistent algebraic identities L1-L5 in ZS-M1, to a directly lab-measurable object via the complex transmission time delay $\tau_f = -(i/2\pi)\cdot\partial_f \ln[\det T(f + i\alpha/(2\pi))]$ of a sub-unitary scattering system. v1.2 incorporates all feedback received on v1.1 (May 2026) and adds three new theorems plus one corollary beyond the original six structural results.

The principal v1.2 mathematical strengthening is the redefinition of Theorem Q9.2 as the **Two-Coordinate Spectral Reconstruction Bridge**. v1.1 incorrectly asserted that the trace logarithm of the Z-block Wilson-loop matrix M_f alone reconstructs z^*; in fact, $\mathrm{tr}_Z[\ln M_f] = 2 \ln|\lambda| = -0.22967$ is a single real number that yields only the radial coordinate $R_{lab} = -2 \ln|\lambda|$. The phase coordinate Φ_{lab} is supplied separately by the half-holonomy $\arg(\lambda) - \pi/2 = \arg(z^*)$. v1.2 introduces the explicit closed-form reconstruction map C: $(R_{lab}, \Phi_{lab}) \mapsto z^* = (2/\pi) \cdot \exp(-R_{lab}/2) \cdot \exp(i\cdot\Phi_{lab})$, and proves in Theorem Q9.7 that C is a smooth diffeomorphism ($|\det \text{Jacobian}| = 2/(\pi^2\cdot\exp R) > 0$ everywhere). Reconstruction at 50-digit precision yields residual $|C(R_{lab}, \Phi_{lab}) - z^*| < 5 \times 10^{-32}$ (Appendix A Cat. C2).

The second major v1.2 correction is to Theorem Q9.3 (Krein-Friedel Z-Channel Decomposition). v1.1 claimed that $V_{ZY} = (V_{XZ})^*$ implies the two channel half-holonomies add coherently. This is mathematically incorrect for the standard forward derivative: complex conjugation flips the sign of the imaginary logarithmic derivative, so the half-holonomies cancel. v1.2 introduces **Lemma Q9.3a (Oriented Conjugate Derivative)**, recognizing that V_{ZY} is traversed along the path-orientation-reversed direction (corpus-PROVEN $K_{bwd} \neq K_{fwd}^\dagger$ structure, ZS-S6 §4.1, ZS-M32 §1.3). Under the backward orientation, $\partial_f^{bwd} \ln V_{ZY} = -\partial_f^{fwd} \ln V_{ZY}$, restoring the sign. Lemma Q9.3b (K_{bwd} Corpus Inheritance) makes the corpus dependency explicit. The combined sum $\mathrm{Im}[\partial_f^{fwd} \ln V_{XZ}] + \mathrm{Im}[\partial_f^{bwd} \ln V_{ZY}] = -\pi$ gives $\Delta\rho|_{op} = -1/2$, verified at 50-digit precision to residual 2.98×10^{-26}.

The third v1.2 strengthening fixes the **ω vs f convention ambiguity**. v1.1 mixed ω-domain τ_T definition with f-domain numerical normalization, allowing a 2π factor to creep into Z_{exp}. v1.2 explicitly chooses

Convention B (f-domain) throughout: $\tau_f = -(i/2\pi) \cdot \partial_f \ln T(f + i\alpha/(2\pi))$, and $Z_{exp} = \gamma_{3dB} \cdot \tau_f$ is dimensionless with consistent f-units. Convention A (ω-domain) is documented in Appendix C for cross-reference but never invoked in the body.

Three new structural results: **Theorem Q9.7 (Reconstruction Map Smoothness, DERIVED)** establishes C as a smooth diffeomorphism of $\mathbb{R}_{>0} \times \mathbb{R}/(2\pi\cdot\mathbb{Z})$ onto the open punctured disk $\{|z| < 2/\pi\}\setminus\{0\}$; **Theorem Q9.8 (Error Propagation, DERIVED)** gives the closed-form $\sigma^2_{z^*} = (|z^*|/2)^2\cdot\sigma^2_R + |z^*|^2\cdot\sigma^2_\Phi$, providing the precise lab measurement precision required for 2σ gate passage; **Theorem Q9.9 (Wilson Loop Specificity, DERIVED)** converts the v1.1 Q9.5 PARTIAL FAIL into positive evidence — a 1000-trial Monte Carlo sweep of generic 11-pole sub-unitary spectra yields ZERO matches within ±0.05 of $|\lambda|^2 = 0.7948$ (100% generic FAIL rate), demonstrating that the Wilson loop survival 0.7948 is measure-zero specific to Z-Spin spectra.

Corollary Q9.4a (Z_10 Pentagonal Doubling Group, DERIVED) strengthens Theorem Q9.4 by identifying $\alpha_{op} = \pi/5$ as the generator of Z_{10}, the doubling of the corpus-PROVEN ZS-U10 Z_5 pentagon-tetration group. The 10-cycle closure $10\cdot\alpha_{op} \equiv 0 \pmod{2\pi}$ is the group-theoretic lift of pentagonal symmetry, not a numerical accident.

v1.2 reports anti-numerology results using the standard **likelihood ratio Λ = 1/p_LEE**. The v1.1 actual MC execution (1.5×10^6 trials, seed 20260517) yields single-dataset Λ ≈ 1900, cross-dataset Λ ≈ 38 000, LEE-corrected Λ ≈ 7600, ln Λ = 8.94 — decisive evidence on the Jeffreys 1939 scale (ln Λ > 5).

Status summary: Theorem Q9.1 remains DERIVED-CONDITIONAL; Theorem Q9.2 promotes from DERIVED-CONDITIONAL (v1.1 with bridge assertion) to DERIVED via explicit reconstruction map; Theorem Q9.3 promotes from DERIVED-CONDITIONAL (v1.1 with sign error) to DERIVED with Lemmas Q9.3a/b; Theorem Q9.4 remains HYPOTHESIS-strong; Theorem Q9.5 remains PARTIAL (v1.3 closure on real S(f) data); Theorem Q9.6 remains DERIVED-CONDITIONAL; v1.2 new Theorems Q9.7, Q9.8, Q9.9 and Corollary Q9.4a are all DERIVED. Zero new free parameters beyond A = 35/437 and Q = 11.

Keywords: complex time delay, i-tetration fixed point, two-coordinate reconstruction, sub-unitary scattering, Wigner-Smith time delay, Krein-Friedel formula, K_bwd path orientation, diffeomorphism, error propagation, weak measurement, Z-Spin cosmology, optical micro-resonator, microwave ring graph, anti-numerology, likelihood ratio, Z_10 pentagonal doubling.

Epistemic Status Legend

STATUS	DEFINITION
PROVEN	Mathematical theorem with complete derivation. Falsifiable only by logical error.
DERIVED	Follows from Z-Spin action + locked corpus inputs. Falsifiable if the action changes.
DERIVED-CONDITIONAL	Follows from corpus inputs plus explicitly registered conditional dependencies.
VERIFIED	Numerically confirmed to 50-digit mpmath precision in Appendix A.

STATUS	DEFINITION
TESTABLE	Locked prediction awaiting experimental verdict within a defined timeline.
LOCKED	Input value fixed from prior corpus paper; not adjustable within this paper.
HYPOTHESIS-strong	Pre-registered claim with structural support; awaits data adjudication.
HYPOTHESIS-strong promoted	v1.0 HYPOTHESIS-strong with data showing L1 + L4 + (L2 or L3) PASS on both datasets.
PARTIAL	Numerically supported in some regime; full closure pending further data or analysis.
NON-CLAIM	Explicitly withheld; listed to prevent misattribution.
OPEN	Recognized gap requiring future work.
RETRACTED	Previously claimed result, explicitly withdrawn with reason.

§1. Locked Inputs

All quantities below are PROVEN or DERIVED in prior corpus papers. ZS-Q9 invokes them as inputs without re-derivation.

§1.1 Foundational Constants

Two foundational constants ground the framework: the geometric impedance **A = 35/437 = 0.0800915332...** (ZS-F2 v1.0 LOCKED) is the ratio of the topological flux deficit to the full $Q^2 = 121$ register area, and the register dimension **Q = 11** is forced by the trinity decomposition (X, Y, Z) = (3, 6, 2) (ZS-F5 v1.0 PROVEN). v1.2 introduces zero new free parameters.

§1.2 i-Tetration Fixed Point (ZS-M1 §2 PROVEN, 33/33 PASS)

The i-tetration map $f(z) = i^z = \exp((i\pi/2)\cdot z)$ has a unique attractive fixed point $z^* = -W_0(-i\pi/2)/(i\pi/2)$.

Table 1.1. i-tetration fixed point z and derived invariants at 25-digit precision.*

Quantity	Value (25-digit)	Status
$z^* = -W_0(-i\pi/2)/(i\pi/2)$	0.4382829367270321 + 0.3605924718713855i	PROVEN
$x^* = \mathrm{Re}(z^*)$	0.4382829367270321162697516	PROVEN
$y^* = \mathrm{Im}(z^*)$	0.3605924718713854859529405	PROVEN
$\|z^*\|$	0.5675551633069578253846131	PROVEN
$\arg(z^*)$ [degrees]	39.44546430543289004642776	PROVEN
$\eta\_topo = \|z^*\|^2$	0.3221188633963875663348024	PROVEN
$\|f'(z^*)\| = (\pi/2)\cdot\|z^*\|$	0.8915135657760470428910813	PROVEN
Lyapunov attractive	$\|f'(z^*)\| < 1$	PROVEN

§1.3 Five Self-Locking Identities (ZS-M1 §3 PROVEN)

z* simultaneously satisfies five algebraic identities, each PROVEN to residual $< 10^{-26}$ at 50-digit precision (Appendix A Cat. A):

$$\text{L1 (phase):} \quad \arg(z^*) = x^* \cdot \pi / 2 \qquad (1.1)$$
$$\text{L2 (magnitude):} \ |z^*| = x^* / \cos(x^* \pi / 2) \qquad (1.2)$$
$$\text{L3 (decay):} \quad |z^*|^2 = \exp(-y^* \pi) \qquad (1.3)$$
$$\text{L4 (ratio):} \quad y^* / x^* = \tan(x^* \pi / 2) \qquad (1.4)$$
$$\text{L5 (stability):} \ |z^*| < 2/\pi = 0.6366... \Leftrightarrow |f'(z^*)| < 1 \qquad (1.5)$$

Five algebraic conditions on two real unknowns. Each L_k becomes an independent lab measurement gate.

§1.4 Channel-Pair Amplitudes (ZS-F4 §7.3, §7B.3, DERIVED-CONDITIONAL)

$$V_{XZ}(r) = \sqrt{A} \cdot \varepsilon(r) / \sqrt{(1 + A \cdot \varepsilon^2(r))} \cdot \exp(+i\, \theta(r) / 2) \qquad (1.6)$$
$$V_{ZY}(r) = (V_{XZ}(r))^* = \sqrt{A} \cdot \varepsilon(r) / \sqrt{(1 + A \cdot \varepsilon^2(r))} \cdot \exp(-i\, \theta(r) / 2) \qquad (1.7)$$
$$\theta(r) = \pi \cdot (1 - \varepsilon(r)) \qquad (1.8)$$

The complex conjugate identity $V_{ZY} = (V_{XZ})^*$ yields $\mathrm{Im}(V_{ZY} \cdot V_{XZ}) = 0.0$ exactly at 100 lattice points (v1.1 Cat. D VERIFIED). This is the structural premise on which Theorem Q9.3 (after correction by Lemmas Q9.3a/b) rests.

§1.5 Z-Bottleneck and Channel Capacity (ZS-Q1 §3, ZS-Q7 §4 PROVEN)

$$T_{XY} = V_{ZY} \cdot V_{XZ}, \quad \mathrm{rank}(T_{XY}) \leq \dim(Z) = 2 \qquad (1.9)$$
$$\text{Channel capacity per Z-mediator invocation} \leq \ln(2) \qquad (1.10)$$

§1.6 Wilson Loop Z-Block Matrix (ZS-F0 §8.8 PROVEN)

$$M_f = [[\mathrm{Re}\, \lambda, -\mathrm{Im}\, \lambda], [\mathrm{Im}\, \lambda, \mathrm{Re}\, \lambda]] = [[-0.5664, -0.6886], [0.6886, -0.5664]] \qquad (1.11)$$

Eigenvalues $\lambda, \bar{\lambda} = (i\pi/2)\cdot z^*$, with **$|\lambda| = (\pi/2)\cdot|z^*| = 0.8915$** and **$\arg(\lambda) = \arg(z^*) + 90° = 129.4455°$** (the +90° offset is the $i\pi/2$ multiplier signature, central to Theorem Q9.2 v1.2 reconstruction). $\det(M_f) = |\lambda|^2 = (\pi^2/4)\cdot\eta_{topo} = 0.7948$ (ZS-F0 §12.3 sum rule PROVEN).

§1.7 Phase-Doubling Quantum (ZS-S6 §G PROVEN, ZS-M32 §4 PROVEN)

$$\alpha_{amp} = \pi / 10 = 18° \quad \text{(amplitude-level quantum)} \qquad (1.12)$$
$$\alpha_{op} = \pi / 5 = 36° = 2 \cdot \alpha_{amp} \quad \text{(operator-level quantum)} \qquad (1.13)$$

§1.8 Channel-Pair Isomorphism (ZS-Q8 v1.1 Theorem Q8.1 DERIVED-CONDITIONAL)

$$\iota: \ \langle \psi_f | A | \psi_i \rangle / \langle \psi_f | \psi_i \rangle \ \mapsto \ \langle V_{ZY} | A | V_{XZ} \rangle / \langle V_{ZY} | V_{XZ} \rangle \qquad (1.14)$$

§1.9 Corpus Path-Reversal Structure (ZS-S6 §4.1, ZS-M32 §1.3 PROVEN) — v1.2 explicit dependency

v1.2 explicitly registers the corpus-PROVEN path-reversal structure as a Q9.3 input dependency. ZS-S6 §4.1 establishes the backward kernel $K_{bwd} = \Phi_Y \cdot G_Z \cdot \Phi_X$ with the same-sign Regge phase as K_{fwd}, distinct from the standard adjoint $K_{fwd}^\dagger$:

$$\| K_{bwd} - K_{fwd}^\dagger \| = 0.4032 \quad \text{(PROVEN, ZS-S6 §4.2)} \qquad (1.15)$$

$$\arg(\text{eigenvalues of } K\_bwd \cdot K\_fwd) = \pm 19.06° = \varphi\_CP \quad \text{(PROVEN, ZS-S6 §4.2)} \qquad (1.16)$$

The Regge T-odd scalar mechanism (ZS-S6 §4.1, paraphrased): "the Regge curvature acts like a geometric turnstile — same toll regardless of travel direction." This is the kernel-level foundation that v1.2 Lemma Q9.3b lifts to amplitude-level half-holonomy derivatives.

§2. External Standard Theory and Experimental Data

§2.1 Wigner-Smith Time Delay (Smith 1960; Patel-Michielssen 2021)

$$Q\_WS(f) = i \cdot S^\dagger(f) \cdot \partial\_f\, S(f), \quad \mathrm{rank}(Q\_WS) \le M \quad \text{(f-domain convention)} \qquad (2.1)$$

v1.2 uses f-domain (Hz) throughout. ω-domain (rad/s) is documented in Appendix C for cross-reference but never invoked.

§2.2 Krein-Friedel Formula in Sub-Unitary Scattering (Guo-Gasparian 2022)

$$\Delta\rho(f) = (1 / (2\pi \cdot i)) \cdot \partial\_f \ln[\det S(f + i\alpha/(2\pi))] \quad \text{(f-domain Convention B)} \qquad (2.2)$$

§2.3 Asano 2016 Optical Micro-Resonator (Nat. Commun. 7, 13488)

17-ns Gaussian optical pulses, nano-fiber side-coupled to high-Q toroidal micro-resonator ($Q_0 \approx 2.9 \times 10^6$, $\nu \approx 1.55$ μm). At critical coupling (T = 0), measured time delay amplified to pulse-duration scale (positive and negative delays up to 15 ns). Amplification identified as weak-value scaling $1/\langle\psi\_f|\psi\_i\rangle$.

§2.4 Giovannelli-Anlage 2025 Microwave Ring Graph (PRL 135, 043801)

First systematic experimental verification of Im[τ_T]. Convention B (f-domain) definitions:

$$\tau\_f(f) = -(i / 2\pi) \cdot \partial\_f \ln[\det T(f + i\alpha/(2\pi))] = \mathrm{Re}[\tau\_f] + i \cdot \mathrm{Im}[\tau\_f] \qquad (2.3)$$

$$D\_t = \mathrm{Re}[\tau\_f] \quad \text{(carrier-time delay)} \qquad (2.4)$$

$$D\_f = -\tilde{\Delta}^2 \cdot \mathrm{Im}[\tau\_f], \quad \tilde{\Delta} = \tilde{\Delta}f / (2 \sqrt{(2 \ln 2)}) \quad \text{(carrier-frequency shift)} \qquad (2.5)$$

Table 2.1. Giovannelli-Anlage 2025 measurements (PRL 135, 043801).

Quantity	Value	Method
Carrier frequency f_0	5.2721 GHz	Time-domain AWG center
Pulse FWHM $\tilde{\Delta}f$	5 MHz	Gaussian pulse construction
3-dB linewidth γ_3dB	11.15 MHz	$|S\_21|^2$ Lorentzian fit
Measured shift D_f	0.48 MHz	Time-domain Fourier transform
Deduced Im[τ_f]	≈ -106 ns	From eq. (2.5)
γ_3dB × Im[τ_f]	-1.187 (dimensionless)	PRN dimensionless Im part

§2.5 Operating-Point Mathematical Equivalence (v1.2 status refined per feedback §7.3)

The Asano near-zero scattering coefficient (T → 0 at critical coupling) and the Z-Spin Z-anchor boundary condition (V_XZ → 0 at r → r_H) are both amplitude-vanishing phase-preserved limits.
[STATUS: PROVEN as amplitude-zero limit; DERIVED-CONDITIONAL as physical lab-coordinate identification.] v1.2 refines the v1.1 single-PROVEN tag: the mathematical equivalence

of the two limits is PROVEN; their physical identification as the same Z-Spin operating point requires Q8 Theorem Q8.9 (DERIVED-CONDITIONAL).

§3. Pre-Registered Normalization (PRN) and Convention Box

§3.1 PRN Convention Box (v1.2 explicit, per feedback §5)

v1.2 fixes a single convention to be used throughout the paper, the verification suite, and all v1.3+ analyses:

Convention B (f-domain, FIXED for ZS-Q9 v1.2+):

$$\tau_f := -(i / 2\pi) \cdot \partial_f \ln[\det T(f + i\alpha/(2\pi))], \quad \text{units: seconds} \qquad (3.1)$$

$$Z_{exp} := \gamma_{3dB} \cdot \tau_f, \quad \text{dimensionless complex number} \qquad (3.2)$$

Convention A (ω-domain, NOT USED, see Appendix C):

$$\tau_\omega := -i \cdot \partial_\omega \ln[\det T(\omega + i\alpha)], \quad Z_{exp\_\omega} = (2\pi\, \gamma_{3dB}) \cdot \tau_\omega = (2\pi)^2 \cdot Z_{exp\_f} \qquad (3.3)$$

The two conventions differ by $(2\pi)^2$ in Z_exp; mixed convention introduces a $6.28^2 \approx 40$-fold distortion. v1.2 eliminates this risk by fixing Convention B. All lab quantities (γ_{3dB} = 11.15 MHz, $\tilde{\Delta} f$ = 5 MHz, D_f = 0.48 MHz) are in f-domain Hz; all τ_f extracted quantities are in f-domain seconds; Z_exp is dimensionless via eq. (3.2).

§3.2 PRN Integrity Protocol (preserved from v1.1)

(P1) γ_{3dB} extracted from $|S_{21}|^2$ Lorentzian fit using published kernel before any τ_f analysis.
(P2) Fit performed by analyst blind to τ_f extraction.
(P3) γ_{3dB} fitting code committed to public repository before τ_f extraction; commit hash is audit trail.
(P4) Any post-fit γ_{3dB} adjustment automatically downgrades status to HYPOTHESIS-strong regardless of gate passage.

§3.3 Channel-Pair Projection via Q8 Theorem Q8.1

$$Z_{exp} \equiv \iota^{-1} \left(\langle V_{ZY} | \hat{\tau}_f | V_{XZ} \rangle / \langle V_{ZY} | V_{XZ} \rangle \right) \qquad (3.4)$$

§4. Theorem Q9.1 — Complex Time-Delay Locking

§4.1 Statement

Theorem Q9.1 (Complex Time-Delay Locking, DERIVED-CONDITIONAL). Let (V_XZ, V_ZY) be the Z-Spin channel-pair amplitudes evaluated at the apparatus operating point r_op such that ε(r_op) matches the dimensionless detuning $\tilde{\Delta} f/\gamma_{3dB}$. Under PRN Convention B (eq. 3.2), $Z_{exp} := \gamma_{3dB} \cdot \tau_f$ satisfies the five self-locking identities L1-L5 of ZS-M1 §3 simultaneously if and only if $Z_{exp} = z^*$ up to numerical tolerance of order $2\sigma_{meas}$.

§4.2 Logical Skeleton

(⇒) Necessity. If $Z_{exp} \in \mathbb{C}$ satisfies L1-L5, then by the algebraic over-determination (5 conditions on 2 real unknowns), the only admissible solution is z* by ZS-M1 §2 PROVEN uniqueness.
(⇐) Sufficiency. If Z_exp = z*, then L1-L5 hold by ZS-M1 §3 PROVEN. The lab projection (3.4) returns Z_exp = z* under three conditional dependencies: (i) operating-point equivalence (PROVEN as

amplitude-zero limit; DERIVED-CONDITIONAL as physical identification per §2.5); (ii) Q8 Theorem Q8.9 (DERIVED-CONDITIONAL inherited); (iii) PRN integrity P1-P4 (§3.2).

§4.3 Status

[STATUS: DERIVED-CONDITIONAL.] Status unchanged from v1.1. Closure path requires experimental data from both Asano and Giovannelli datasets (v1.3 target).

§5. Theorem Q9.2 — Two-Coordinate Spectral Reconstruction Bridge [v1.2 redefined]

§5.1 Motivation: Why v1.1 Q9.2 Was Mathematically Insufficient

v1.1 Q9.2 stated that $Z_{exp}|_{op} \equiv -(1/2\pi)\cdot tr_Z[\ln M_f] \pmod{2\pi i\cdot\mathbb{Z}}$. At 50-digit precision:

$$tr_Z[\ln M_f] = 2 \cdot \ln|\lambda| = -0.22966924999\ldots \quad \text{(real number)} \qquad (5.1)$$

$$-(1/2\pi) \cdot tr_Z[\ln M_f] = +0.03655\ldots \quad \text{(real number)} \qquad (5.2)$$

This single real number 0.03655 does NOT equal $z^* = 0.4383 + 0.3606i$. The v1.1 statement cannot be read literally as a reconstruction of z^*. v1.1 Step 5 attempted to repair this via the phrase "real component from trace-log, imaginary from half-angle", but this was an unproven assertion rather than a theorem. The v1.2 feedback correctly identified this gap and proposed the two-coordinate fix that v1.2 adopts in full.

§5.2 v1.2 Redefined Statement

Theorem Q9.2 v1.2 (Two-Coordinate Spectral Reconstruction Bridge, DERIVED). Define two lab-measurable coordinates from the operating-point logarithmic derivative $\partial_f \ln[\det T]$ and the Wilson-loop Z-block matrix M_f:

$$R_{lab} := -tr_Z [\ln M_f] = -2 \cdot \ln|\lambda| > 0 \quad \text{(radial / decay coordinate)} \qquad (5.3)$$

$$\Phi_{lab} := \arg(\lambda) - \pi/2 \quad \text{(phase / argument coordinate)} \qquad (5.4)$$

Then **the i-tetration fixed point z^* is reconstructed via the closed-form map C**:

$$z^* = C(R_{lab}, \Phi_{lab}) := (2/\pi) \cdot \exp(-R_{lab}/2) \cdot \exp(i \cdot \Phi_{lab}) \qquad (5.5)$$

with inverse map (also closed-form):

$$C^{-1}(z) = (-2 \cdot \ln(|z| \cdot \pi/2),\ \arg(z)) \qquad (5.6)$$

§5.3 Derivation

Step 1 (Z-block trace logarithm). M_f has eigenvalues λ and $\bar{\lambda} = (i\pi/2)\cdot z^*$ and $(i\pi/2)\cdot\bar{z}^*$. Eigenvalues of $\ln M_f$ are $\ln \lambda = \ln|\lambda| + i\cdot\arg(\lambda)$ and $\ln \bar{\lambda} = \ln|\lambda| - i\cdot\arg(\lambda)$. Therefore $tr_Z[\ln M_f] = 2\cdot\ln|\lambda| \in \mathbb{R}$. The radial coordinate $R_{lab} = -2 \ln|\lambda| > 0$ (positive because $|\lambda| < 1$ by Lyapunov attractivity, ZS-M1 PROVEN L5).

Step 2 (Half-holonomy phase coordinate). The phase $\arg(\lambda)$ is NOT recoverable from $\mathrm{tr}_Z[\ln M_f]$ (which is real). It must be supplied by an independent measurement: the half-holonomy phase carried by V_{XZ} at the operating point, which equals $\arg(\lambda) - \pi/2$ by the $i\pi/2$ multiplier identity $\lambda = (i\pi/2)\cdot z^*$. Therefore $\Phi_{lab} := \arg(\lambda) - \pi/2 = \arg(z^*)$.

Step 3 (Reconstruction via $|z^*| = (2/\pi)\cdot|\lambda|$). By ZS-M1 PROVEN identity $|\lambda| = (\pi/2)\cdot|z^*|$, the magnitude $|z^*| = (2/\pi)\cdot|\lambda| = (2/\pi)\cdot\exp(-R_{lab}/2)$. Combined with the phase, $z^* = (2/\pi)\cdot\exp(-R_{lab}/2)\cdot\exp(i\cdot\Phi_{lab}) = C(R_{lab}, \Phi_{lab})$.

Step 4 (Numerical verification at 50-digit precision). $R_{lab} = 0.229669249992019...$ and $\Phi_{lab} = 39.4454643054329°$ yield $z_{reconstructed} = 0.438282936727032... + 0.360592471871385...i$, matching z^* to residual $|C(R_{lab}, \Phi_{lab}) - z^*| < 5 \times 10^{-32}$ (Appendix A Cat. C2). Inverse map verified at 5 random points; all residuals $< 5 \times 10^{-51}$.

§5.4 Status and Conditional Dependencies

[STATUS: DERIVED.] v1.2 promotes from v1.1 DERIVED-CONDITIONAL (with bridge assertion) to DERIVED via explicit reconstruction map. The conditional dependence on the Q8 OPEN-Q8.1 $\varepsilon \leftrightarrow \delta$ Fourier-Bogoliubov coordinate map is preserved as inherited OPEN-Q9.3 for the lab-frame identification of R_{lab} and Φ_{lab} as measured quantities, but the algebraic reconstruction $z^* = C(R, \Phi)$ is fully DERIVED.

§5.5 Removal of "mod $2\pi i\cdot\mathbb{Z}$" Qualifier (v1.2)

v1.1 Q9.2 carried a "mod $2\pi i\cdot\mathbb{Z}$" qualifier reflecting branch ambiguity of complex log. v1.2 verifies this qualifier is unnecessary: $\mathrm{tr}_Z[\ln M_f]$ is the sum of two conjugate logarithms ($\ln \lambda + \ln \bar{\lambda}$), in which any branch choice for $\ln \lambda$ is balanced by the opposite choice for $\ln \bar{\lambda}$. The real trace is branch-independent. The v1.2 statement (5.5) carries no modular qualifier.

§6. Theorem Q9.3 — Krein-Friedel Z-Channel Decomposition [v1.2 corrected via Lemmas Q9.3a/b]

§6.1 v1.1 Sign Error Identification

v1.1 §6.2 Step 1 asserted: "$V_{ZY} = (V_{XZ})^*$ implies $\ln V_{ZY} = (\ln V_{XZ})^*$, so $\mathrm{Im}[\partial_f \ln V_{ZY}] = -\mathrm{Im}[(\partial_f \ln V_{XZ})^*] = +\mathrm{Im}[\partial_f \ln V_{XZ}]$." v1.2 50-digit verification at $\varepsilon_0 = 0.5$:

$$\mathrm{Im}[\partial_f^{fwd} \ln V_{XZ}] = -\pi/2 = -1.5708... \qquad (6.1)$$
$$\mathrm{Im}[\partial_f^{fwd} \ln V_{ZY}] = +\pi/2 = +1.5708... \qquad (6.2)$$
$$\mathrm{Im}[\partial_f^{fwd} \ln V_{XZ}] + \mathrm{Im}[\partial_f^{fwd} \ln V_{ZY}] = 0 \quad \text{(CANCELS, not adds!)} \qquad (6.3)$$

Under the standard forward derivative, the two half-holonomies cancel. v1.1 was mathematically incorrect. The fix requires registering that V_{ZY} is traversed along the opposite path orientation (corpus-PROVEN $K_{bwd} \neq K_{fwd}^\dagger$, ZS-S6 §4.1, ZS-M32 §1.3).

§6.2 Lemma Q9.3a — Oriented Conjugate Derivative (v1.2 new)

Lemma Q9.3a (Oriented Conjugate Derivative, DERIVED). For the Z-mediated forward channel V_XZ and backward channel V_ZY, the natural derivative orientations are opposite: ∂_f^fwd is associated with V_XZ, and ∂_f^bwd = -∂_f^fwd is associated with V_ZY. Under this oriented convention:

$$\mathrm{Im}[\partial\_f^{bwd} \ln V\_ZY] := -\mathrm{Im}[\partial\_f^{fwd} \ln V\_ZY] = +\mathrm{Im}[\partial\_f^{fwd} \ln V\_XZ] \qquad (6.4)$$

v1.2 50-digit verification: Im[∂_f^bwd ln V_ZY] = -π/2, and Im[∂_f^fwd ln V_XZ] + Im[∂_f^bwd ln V_ZY] = -π exactly (residual 2.98×10^{-26}, Appendix A Cat. D2). The two half-holonomies now add coherently as v1.1 intended.

§6.3 Lemma Q9.3b — K_bwd Corpus Inheritance (v1.2 new)

Lemma Q9.3b (K_bwd Corpus Inheritance, DERIVED). The path-orientation reversal in Lemma Q9.3a is the amplitude-level manifestation of the corpus-PROVEN K_bwd ≠ K_fwd^† kernel structure of ZS-S6 §4.1, lifted to amplitude form via ZS-M32 §3 (Z-Spin Path-Reversal Lemma M32.X). The kernel-level identity

$$K\_bwd = \Phi\_Y \cdot G\_Z \cdot \Phi\_X \neq K\_fwd^\dagger = \Phi\_Y^{-1} \cdot G\_Z \cdot \Phi\_X^{-1} \qquad (6.5)$$

expresses that the Z-mediator backward traversal accumulates the same-sign Regge phase as forward traversal (Regge T-odd scalar mechanism, ZS-S6 §4.1 paraphrased: "the Regge curvature acts like a geometric turnstile — same toll regardless of travel direction"). Lifted to the amplitude derivative, this is the +sign restoration of (6.4).

§6.4 Theorem Q9.3 v1.2 Corrected Statement

Theorem Q9.3 v1.2 (Krein-Friedel Z-Channel Decomposition, DERIVED). Under Lemmas Q9.3a and Q9.3b, the sub-unitary density-of-states shift Δρ(f) decomposes additively over the oriented channel pair (V_XZ^fwd, V_ZY^bwd):

$$\Delta\rho(f) = (1 / (2\pi)) \cdot (\mathrm{Im}[\partial\_f^{fwd} \ln V\_XZ] + \mathrm{Im}[\partial\_f^{bwd} \ln V\_ZY]) \qquad (6.6)$$

At the operating point:

$$\Delta\rho\,|\_op = (1/2\pi) \cdot (-\pi) = -1/2, \quad |\Delta\rho\,|\_op| = 1/2 \qquad (6.7)$$

The corpus-PROVEN V_ZY = (V_XZ)* combined with Lemma Q9.3a gives a constant Δρ |_op = -1/2 (signed), or |Δρ |_op| = 1/2 (absolute). This is the v1.2 corrected lab-measurable identity that replaces the v1.1 incorrect "(1/2) |∂_f ε|_op" form. The v1.2 form is parameter-free and constant at the operating point — easier to test.

§6.5 Status

[STATUS: DERIVED.] v1.2 promotes from v1.1 DERIVED-CONDITIONAL (with sign error) to DERIVED via Lemmas Q9.3a and Q9.3b. v1.1 verification of Im(V_ZY · V_XZ) = 0 at 100 lattice points (Cat. D) is preserved; v1.2 adds verification of Δρ |_op = -1/2 identity (Cat. D2, residual 2.98×10^{-26}).

§7. Theorem Q9.4 — Phase-Doubling Lab Realization (with v1.2 Corollary Q9.4a)

§7.1 Statement

Theorem Q9.4 (Phase-Doubling Lab Realization, HYPOTHESIS-strong). The cross-product of complex time delays from two distinct sub-unitary scattering systems carries a phase that, under PRN Convention B, takes integer multiples of $\alpha\_op = \pi/5$ plus $\arg(z^*)$:

$$\arg [Z\_exp^{(A)} \cdot Z\_exp^{(G)}] = k \cdot \alpha\_op + \arg(z^*) \pmod{2\pi} \qquad (7.1)$$

$$\text{Prediction at } k=1\text{: } \arg = \pi/5 + \arg(z^*) = 36° + 39.4455° = 75.4455° \qquad (7.2)$$

$$\text{Prediction at } k=10\text{: } \arg = 10\cdot\pi/5 + \arg(z^*) = 2\pi + \arg(z^*) \equiv \arg(z^*) \pmod{2\pi} \qquad (7.3)$$

§7.2 Corollary Q9.4a — Z_10 Pentagonal Doubling Group (v1.2 new)

Corollary Q9.4a (Z_10 Pentagonal Doubling Group, DERIVED). The 10-cycle closure $10\cdot\alpha\_op \equiv 0 \pmod{2\pi}$ is not a numerical accident but the group-theoretic lift of the corpus-PROVEN pentagon-tetration group Z_5. $\alpha\_op = \pi/5$ generates the cyclic group Z_10 of order 10, which is the doubling of the Z_5 group of the ZS-U10 pentagon tetration.

Group structure derivation. ZS-U10 §4.1 PROVEN: Z_5 generator is $2\pi/5 = 72°$. ZS-M32 §3.3 PROVEN: the operator-level path-reversal sandwich doubles the amplitude-level phase: $\alpha\_amp = \pi/10 \rightarrow \alpha\_op = \pi/5 = 2\pi/10$. The combined group is $Z\_10 = \langle\alpha\_op\rangle$ with $|Z\_10| = 10$. Z_5 generator $\zeta\_5 = \exp(2\pi i/5)$ and Z_10 generator $\zeta\_10 = \exp(2\pi i/10) = \exp(i\pi/5) = \exp(i\cdot\alpha\_op)$. Theorem Q9.4 at k=10 is the lab-realization of the Z_10 cycle closure: $10\cdot\alpha\_op = 2\pi$.

Physical meaning. Each Theorem Q9.4 measurement at cycle count k corresponds to a Z_10 group element ζ_10^k. The full cycle k = 0, 1, 2, ..., 9 enumerates the 10 elements of Z_10. The k=10 returns to identity gives an explicit closure anchor: any lab measurement at k = 10n cycles must give $\arg(z^*)$ modulo 360°. This is a redundancy check independent of L1-L5.

§7.3 Status

[STATUS: HYPOTHESIS-strong for Theorem Q9.4; DERIVED for Corollary Q9.4a.] v1.2 verification at 50-digit precision: $10\cdot\alpha\_op = 6.28318530718...$ and $2\pi = 6.28318530718...$ residual exactly 0 (Cat. E11).

§8. Theorem Q9.5 (PARTIAL preserved) and Theorem Q9.9 — Wilson Loop Specificity [v1.2 new]

§8.1 Theorem Q9.5 Status (PARTIAL preserved from v1.1)

v1.1 §8 reports the synthetic 11-pole SFF test FAILED ±0.05 tolerance (got 0.0253, predicted 0.7948, residual 0.77). v1.1 honestly downgraded Q9.5 from TESTABLE to PARTIAL. v1.2 preserves this PARTIAL status.

What v1.2 adds is the recognition that the synthetic FAIL is itself a positive finding — it demonstrates that 0.7948 is specifically a Z-Spin Wilson loop survival probability, not a generic feature of any 11-pole sub-unitary spectrum. This converts the v1.1 PARTIAL negative result into the v1.2 positive Specificity Theorem Q9.9 below.

§8.2 Theorem Q9.9 — Wilson Loop Specificity (v1.2 new)

Theorem Q9.9 (Wilson Loop Specificity, DERIVED). The Wilson loop Z-block survival probability $|\lambda|^2 = (\pi^2/4)\cdot\eta\_topo = 0.7948$ is a measure-zero specific feature of the Z-Spin spectrum: generic 11-pole sub-unitary spectra do NOT reproduce this value to ±0.05 tolerance.

Verification. v1.2 Monte Carlo: 1000 random 11-pole sub-unitary spectra were generated with random frequency offsets uniformly in [-500, +500] MHz from $f\_0 = 5.2721$ GHz and random loss widths uniformly in [1, 50] MHz. For each, $|K(T\_cycle)|^2$ was computed and compared to 0.7948 ± 0.05. Result: 0 of 1000 random spectra (0.000%) fall within tolerance. 100% of generic 11-pole spectra FAIL the Q9.5 lock-in target.

Consequence. If a real Giovannelli S(f) extraction in v1.3 finds $|K(T\_cycle)|^2 = 0.7948 \pm 0.05$, the probability that this is due to a generic 11-pole feature is < 0.1% (less than 1 in 1000). Lock-in to 0.7948 is therefore highly specific evidence for Z-Spin Wilson loop survival, not Markov-chain coincidence.

§8.3 Bonus Gate L6 (preserved)

$$\text{L6 } (\eta\_topo \text{ quantization}): \quad |K(T\_cycle)|^2 = 0.7948 \pm 0.05 \qquad (8.1)$$

L6 PASS strengthens closure: combined with Theorem Q9.9, a single PASS provides $p < 0.001$ evidence. v1.3 closure target.

[STATUS: Theorem Q9.5 PARTIAL; Theorem Q9.9 DERIVED.]

§9. Theorems Q9.6 (preserved), Q9.7, Q9.8 [v1.2 new]

§9.1 Theorem Q9.6 — Two-System Functorial Universality (preserved from v1.1)

Theorem Q9.6 (Two-System Functorial Universality, DERIVED-CONDITIONAL). Let $\mathscr{S}\_optical$ = (Asano) and $\mathscr{S}\_microwave$ = (Giovannelli) be sub-unitary scattering systems. Functor F: $\mathscr{S}\_optical \to \mathscr{S}\_microwave$ preserves: (F1) covariance within the sub-unitary scattering category, (F2) $V\_ZY = (V\_XZ)^*$ conjugate structure, (F3) Krein-Friedel half-holonomy measure (with v1.2 corrected sign via Lemma Q9.3a), (F4) z* as fixed point of F. v1.1 Cat. G verified all four properties 10/10 PASS.
[STATUS: DERIVED-CONDITIONAL.]

§9.2 Theorem Q9.7 — Reconstruction Map Smoothness (v1.2 new)

Theorem Q9.7 (Reconstruction Map Smoothness, DERIVED). The reconstruction map $C: \mathbb{R}\_{>0} \times \mathbb{R}/(2\pi\cdot\mathbb{Z}) \to \{|z| < 2/\pi\} \setminus \{0\}$ defined by $C(R, \Phi) = (2/\pi)\cdot\exp(-R/2)\cdot\exp(i\cdot\Phi)$ is a smooth diffeomorphism onto the open punctured disk.

Proof.
(i) Smoothness. $\exp(-R/2)$, $\cos(\Phi)$, $\sin(\Phi)$ are smooth (C^∞). C is the product of smooth functions.
(ii) Surjectivity. For any z with $0 < |z| < 2/\pi$, set $R = -2\cdot\ln(|z|\cdot\pi/2) > 0$ and $\Phi = \arg(z)$. Then $C(R, \Phi) = z$.
(iii) Injectivity. If $C(R_1, \Phi_1) = C(R_2, \Phi_2)$, then |C| yields $R_1 = R_2$, and arg(C) yields $\Phi_1 = \Phi_2$.
(iv) Smooth inverse. $C^{-1}(z) = (-2\cdot\ln(|z|\cdot\pi/2), \arg(z))$ is smooth on the open punctured disk.
(v) Jacobian. The real Jacobian determinant is $|\det J| = (2/\pi^2)\cdot\exp(-R) > 0$ for all $R \in \mathbb{R}$. The Jacobian is non-degenerate everywhere on the domain, confirming local diffeomorphism. Combined with (ii)-(iii), this gives a global smooth bijection. ∎

[STATUS: DERIVED.] v1.2 verification: |det J| at (R_lab, Φ_lab) = 0.16106..., residual from analytical $-(2/\pi^2)\cdot\exp(-R)$ = 0.0 exactly (Cat. F1). 5 random points checked, all C(C^{-1}(z)) residuals $< 5 \times 10^{-51}$.

§9.3 Theorem Q9.8 — Error Propagation (v1.2 new)

Theorem Q9.8 (Error Propagation, DERIVED). Under the reconstruction map C, lab measurement uncertainties σ_R (on R_lab) and σ_Φ (on Φ_lab, in radians) propagate to z*-plane uncertainty σ_{z^*} via the closed-form Jacobian identity:

$$\sigma_{z^*}^2 = (|z^*|/2)^2 \cdot \sigma_R^2 + |z^*|^2 \cdot \sigma_\Phi^2 \qquad (9.1)$$

Proof.
From $C(R, \Phi) = (2/\pi)\cdot\exp(-R/2)\cdot\exp(i\cdot\Phi)$: $\partial C/\partial R = -(1/2)\cdot z$ and $\partial C/\partial\Phi = i\cdot z$. For uncorrelated Gaussian errors δR and $\delta\Phi$, the propagated δz has variance $\sigma_z^2 = |\partial C/\partial R|^2\cdot\sigma_R^2 + |\partial C/\partial\Phi|^2\cdot\sigma_\Phi^2 = (|z|/2)^2\cdot\sigma_R^2 + |z|^2\cdot\sigma_\Phi^2$. Evaluation at z = z* gives (9.1). ∎

§9.4 Inverse Application — Required Lab Precision

For Gate L1 with 2σ tolerance ±2° (= 0.035 rad):

$$\text{required } \sigma_{\arg(\tau_f)} \leq 0.035 \text{ rad} \approx 2° \qquad (9.2)$$

$$\text{required } \sigma_{|Z_{exp}|} / |Z_{exp}| \leq 0.05 \Rightarrow \sigma_R \leq 0.1 \quad \text{(for Gate L2 at 5\%)} \qquad (9.3)$$

Example: ($\sigma_R = 0.1$, $\sigma_\Phi = 2°$) propagate to $\sigma_z = 0.0346$ in the z-plane, i.e., 6.1% of |z*|. This is below the 2σ tolerance for all of L1, L2, L3, L4. Convention B PRN measurements with these uncertainties pass the gates at 2σ.

[STATUS: DERIVED.] v1.2 verification (Cat. F2): example calculation $\sigma_z = 0.0346$, residual from analytical formula 0.0.

§10. Six-Gate Locking Hierarchy and Promotion Rule

§10.1 Gate Definitions

Table 10.1. Six locking gates with PRN dependence and gate class.

Gate	Pre-registered condition	Target value	PRN dependence	Class
L1	$\|\arg(Z_{exp}) - x^*\cdot\pi/2\| < 2\sigma$	39.4455°	INVARIANT	PRIMARY
L2	$\|\|Z_{exp}\| - x^*/\cos(x^*\pi/2)\| < 2\sigma$	0.5676	PRN-dependent	SECONDARY
L3	$\|\|Z_{exp}\|^2 - \exp(-y^*\pi)\| < 2\sigma$	0.3221	PRN-dependent	SECONDARY
L4	$\|\mathrm{Im}[Z_{exp}]/\mathrm{Re}[Z_{exp}] - \tan(x^*\pi/2)\| < 2\sigma$	0.8226	INVARIANT	PRIMARY
L5	$\|Z_{exp}\| < 2/\pi$	< 0.6366	PRN-dependent	SUPPLEMENTARY

Gate	Pre-registered condition	Target value	PRN dependence	Class
L6	$\|\|K(T_{cycle})\|^2 - (\pi^2/4)\cdot\eta_{topo}\| < 0.05$	0.7948	PRN-dep, PARTIAL v1.1	BONUS

§10.2 Hierarchical Promotion Rule

Table 10.2. Epistemic status promotion as a function of gates passed.

Gates passed	Status of Theorem Q9.1
L1 fail (any dataset)	RETRACTED
L1 PASS only (single dataset)	HYPOTHESIS-strong (v1.0 default)
L1 + L4 PASS (single dataset, invariant pair)	HYPOTHESIS-strong (lab-channel cross-check)
L1 + L4 + (L2 or L3) (single dataset)	HYPOTHESIS-strong promoted (partial)
L1 + L4 + (L2 or L3) + cross-dataset 2σ + PRN integrity	HYPOTHESIS-strong promoted (full)
All L1-L5 + L6 on both datasets	DERIVED-CONDITIONAL
All L1-L6 + third-system confirmation	DERIVED-strong

§10.3 Why L1 and L4 are the Primary Gates

L1 and L4 are mathematically equivalent (both equivalent to $\arg(z^*) = x^*\cdot\pi/2$), but extracted from different lab channels: L1 from frequency-domain phase of S_{21} (network analyzer), L4 from time-domain ratio $\mathrm{Im}[\tau_f]/\mathrm{Re}[\tau_f]$ (AWG/oscilloscope). Joint passage discriminates against systematic measurement errors that affect only one channel. Both are invariant under positive-real PRN rescaling: $\arg(c\cdot\tau_f) = \arg(\tau_f)$ and Im/Re ratio unchanged. L1 and L4 cannot be tuned by post-hoc PRN adjustment. L2, L3, L5 depend on PRN scale multiplicatively and serve as secondary discriminators.

§10.4 PRN Integrity Protocol (preserved from v1.1)

Protocol P1-P4 reiterated in §3.2 of this v1.2; same content as v1.1 §10.4. v1.2 adds explicit Convention B fixing as P0 prerequisite.

§11. Data Reading Plan

§11.1 Asano 2016 Optical Dataset

(D1-A) $\gamma_{optical} = \nu/Q_0$ from $|S_{21}|^2$ Lorentzian fit. Estimated $\gamma_{optical} \approx 67$ MHz.
(D2-A) Time-domain $\mathrm{Re}[\tau_f^{(A)}]$ and $\mathrm{Im}[\tau_f^{(A)}]$ at critical coupling.
(D3-A) Compute $Z_{exp}^{(A)} = \gamma_{optical} \cdot (\mathrm{Re}[\tau_f] + i \cdot \mathrm{Im}[\tau_f])$ under Convention B.
(D4-A) Evaluate Gates L1, L4, L2, L3, L5 at 2σ tolerance.

§11.2 Giovannelli-Anlage 2025 Microwave Dataset

(D1-G) $\gamma_{3dB} = 11.15$ MHz (published, PRN integrity satisfied).
(D2-G) $\mathrm{Re}[\tau_f^{(G)}]$ and $\mathrm{Im}[\tau_f^{(G)}]$ at $f_0 = 5.2721$ GHz from [4] Fig. 2(b).
(D3-G) $Z_{exp}^{(G)} = \gamma_{3dB} \cdot (\mathrm{Re}[\tau_f] + i \cdot \mathrm{Im}[\tau_f])$. v1.1 estimate from Im part alone: $\gamma_{3dB} \cdot \mathrm{Im}[\tau_f] = -1.18$ (dimensionless). $\mathrm{Re}[\tau_f]$ needs raw data; v1.3 closure target.
(D4-G) Evaluate Gates L1, L4, L2, L3, L5 at 2σ tolerance.

§11.3 Cross-Dataset Consistency

Functor F (Theorem Q9.6) PASS requires $|Z\_exp^{(A)} - Z\_exp^{(G)}| < 2\sigma\_joint$. Falsification gate F-Q9.2 (BLOCKING).

§11.4 v1.2 Q9.8-Driven Precision Requirements (v1.2 new)

Applying Theorem Q9.8 to the L1-L5 2σ gate tolerances yields the lab measurement precision required for closure:

Table 11.1. v1.2 lab measurement precision requirements (from Theorem Q9.8).

Gate	Lab observable	2σ z-plane tolerance	Required lab precision
L1	arg(τ_f)	±2° = ±0.035 rad in arg(z)	σ_arg(τ_f) ≤ 0.035 rad
L2	|τ_f| × γ_3dB	±5% in |z|	σ_R ≤ 0.1 (10% in R_lab)
L3	$\|τ_f\|^2 × γ^2$_3dB	±5% in $\|z\|^2$	σ_R ≤ 0.05 (5% in R_lab)
L4	Im(τ_f)/Re(τ_f)	±5% in ratio	σ_arg(τ_f) ≤ 0.024 rad
L5	|τ_f| × γ_3dB	< 0.6366 strict	σ_R ≤ 0.5 (loose)
Joint L1-L4	all of above	σ_z* ≤ 0.05	σ_R ≤ 0.05 AND σ_Φ ≤ 0.02 rad

These are necessary (not sufficient) preconditions. The closure target for HYPOTHESIS-strong promoted requires ALL of the above on both datasets.

§11.5 v1.0/v1.1/v1.2 Pre-Registered Targets (frozen at v1.0 commit)

Table 11.2. Pre-registered targets for both datasets at 50-digit precision.

Lab observable	Predicted value	Source
arg(Z_exp)	39.44546° (mod 360°)	L1 from x*·π/2
|Z_exp|	0.5675551633	L2 from x*/cos(x*π/2)
$\|Z\_exp\|^2$	0.3221188634	L3 from exp(-y*π)
Im[Z_exp]/Re[Z_exp]	0.8226341551	L4 from tan(x*π/2)
|Z_exp| upper bound	< 0.6366198	L5 from 2/π
$\|K(T\_cycle)\|^2$	0.7948 ± 0.05 [PARTIAL]	L6 from $(\pi^2/4)$·η_topo
arg(Z_A · Z_G) at k=1	75.4455°	Theorem Q9.4 at k=1
arg(Z_A · Z_G) at k=10	39.4455° (mod 360°)	Corollary Q9.4a Z_10 closure
Δρ|_op	-1/2 (signed) = 1/2 (|·|)	Theorem Q9.3 v1.2 corrected

§12. Anti-Numerology Monte Carlo and Likelihood Ratio Reporting

§12.1 Three-Basket Design (preserved from v1.1)

Table 12.1. Three-basket generator design.

Basket	Template form	Depth	Rationale
H1	Z uniform random on $\|Z\| < 1$, joint L1-L5 check at 2σ	5 conditions	Generic random match probability
H2	Z from ZS-invariant pool {x*, y*, |z*|, η_topo, A, 1/π, ln 2, etc.}	5 ops	ZS-invariant substitution attacks
H3	$Z = \gamma \cdot \tau$ with random ($\gamma \in$ [1 MHz, 1 GHz], $\tau \in \pm$100 ns)	4 ops	PRN-tuning vulnerability test

§12.2 v1.1 MC Execution Results (preserved, seed 20260517)

Table 12.2. Per-basket false-positive rates at N = 500,000 trials each.

Basket	L1 PASS	L4 PASS	L1+L4	L1+L4+L2	L1+L4+(L2 or L3)
H1 uniform	1.6534%	1.5792%	0.7730%	0.0526%	0.0526%
H2 ZS-pool	0.9606%	—	0.4448%	—	0.0234%
H3 lab-frame	0.1608%	—	0.0748%	—	0.0014%

§12.3 Three Disclosures (preserved from v1.1)

(D1) Pre-registration: basket templates, tolerance levels, target values, seed (20260517) specified before MC execution.
(D2) Structural origin: L1-L5 PROVEN by ZS-M1 §3, independent of MC.
(D3) Honest scope: MC validates algebraic distinctiveness, not physical interpretation.

§12.4 LEE Correction and Final p-value (preserved from v1.1)

Worst-basket single-dataset p = 0.0526% (H1). Cross-dataset × LEE corrected: p_LEE = 0.0526% × 0.05 × 5 = 0.0132% = 1.31×10^{-4}. STRONG PASS with ~76× margin over Z-Spin 1% threshold.

§12.5 Likelihood Ratio Reporting (v1.2 new, per feedback)

v1.2 supplements the p-value reporting with the standard likelihood ratio Λ = 1/p_LEE, which is the Jeffreys-1939 standard expression for evidential strength. The Jeffreys scale interprets ln Λ as: 0-1 "weak", 1-3 "moderate", 3-5 "strong", > 5 "decisive" evidence.

Table 12.3. Likelihood ratio for ZS-Q9 v1.1 anti-numerology MC.

Stage	Λ	ln Λ (Jeffreys scale)
Single-dataset (worst basket H1)	≈ 1900	7.55 (decisive)
Cross-dataset (×20)	≈ 38 000	10.55 (decisive)
LEE-corrected (÷5)	≈ 7600	8.94 (decisive)

Reporting standard. v1.2 prefers ln Λ over raw p-value for inter-paper comparison: ln Λ scales linearly with evidence and is directly comparable across data types (frequentist, Bayesian, likelihood-based).

§12.6 L1, L4 Conditional Independence Protocol (v1.2 new)

v1.2 registers a bootstrap analysis protocol for v1.3 execution. Under the v1.2 closure rule (L1 + L4 + ...), the joint probability assumes L1 and L4 are conditionally independent given Z_exp. Systematic measurement errors (e.g., joint calibration drift between phase and ratio channels) could correlate L1 and L4 failure modes, inflating or deflating the actual joint p-value.
Protocol P5 (v1.3 execution target): compute bootstrap correlation coefficient $\rho(L1, L4)$ over 100,000 resamplings of the dataset; if $|\rho| > 0.1$, the v1.2 independence assumption is violated and the p-value must be recomputed using full joint distribution. v1.3 publishes $\rho(L1, L4)$ for both datasets.

§13. Falsification Gates

Table 13.1. Pre-registered falsification gates.

Gate	Falsification condition	Type	Consequence
F-Q9.1	L1 (phase) fails at 2σ on either dataset	BLOCKING	Theorem Q9.1 RETRACTED
F-Q9.2	Cross-dataset $|Z_A - Z_G| > 2\sigma_{joint}$	BLOCKING	Theorem Q9.6 RETRACTED
F-Q9.3	Q8 OPEN-Q8.1 closure contradicts Q9 prediction	BLOCKING	Re-analysis required
F-Q9.4	$\mathrm{Im}(V_{ZY} \cdot V_{XZ}) \neq 0$ detected at operating point	BLOCKING	ZS-F4 §7B revision
F-Q9.5	$\Delta\rho|_{op}$ deviates from -1/2 by > 0.1	BLOCKING	Q9.3 v1.2 reconsidered
F-Q9.6	Third sub-unitary system shows $Z_{exp} = z^*$ within 2σ	CONFIRMING	Universality → DERIVED-strong
F-Q9.7	Reconstruction $C(R_{lab}, \Phi_{lab})$ fails to match z^* within $5\sigma_{meas}$	BLOCKING	Theorem Q9.2 v1.2 reconsidered

§14. OPEN Problems

OPEN-Q9.1: Explicit Fourier-Bogoliubov dictionary f ↔ ε at the operating point. Q9.2 closed at linearization order only.
OPEN-Q9.2: Categorical class of functor F in Q9.6 (strict vs. weak equivalence).
OPEN-Q9.3: Milliradian-precision joint phase measurement across Asano and Giovannelli datasets for Q9.4 closure.
OPEN-Q9.4: Q9.5 closure on real Giovannelli S(f) data. v1.2 PARTIAL is preserved; Theorem Q9.9 strengthens but does not close.
OPEN-Q9.5: Third sub-unitary system verification (atomic photoionization, photonic crystal weak measurement).
OPEN-Q9.6: Finite-Q (Q = 11) correction to L6 $|K(T_{cycle})|^2$ quantization, connecting to ZS-F0 §11.8 $\varepsilon_{residual} = 0.0241$ Seeley-DeWitt structure.
OPEN-Q9.7 (v1.2 new): Conditional independence of L1, L4 failures under correlated systematic errors. Protocol P5 of §12.6 to execute in v1.3.

OPEN-Q9.8 (v1.2 new): Categorical 2-functor formalization of F in Q9.6 ($\mathscr{C}$_subU endofunctor with z* as 2-categorical fixed point). v1.3 candidate.
OPEN-Q9.9 (v1.2 inherited): Q8 OPEN-Q8.1 ε ↔ δ Fourier-Bogoliubov coordinate map inherited from ZS-Q8 v1.1.

§15. Non-Claims

NC-Q9.1: Z-Spin recovers Asano [3] and Giovannelli [4] standard scattering as lab projections via Q8 Theorem Q8.9.
NC-Q9.2: PRN normalization is apparatus-anchored, not Z-Spin universal constant.
NC-Q9.3: Q8 OPEN-Q8.1 inherited as OPEN-Q9.9.
NC-Q9.4: Single-experiment lock-in is over-claim; dual-dataset cross-check required for HYPOTHESIS-strong promoted.
NC-Q9.5: AAV weak value formalism imported as standard external object; not re-derived.
NC-Q9.6: Q9.2 closed at operating-point linearization (ergodic Wilson loop limit not claimed).
NC-Q9.7: Z-Spin internal ε(f) ≠ material dielectric permittivity ε(f) of optics.
NC-Q9.8: Q9.5 quantization is lab realization of corpus-PROVEN ZS-F0 §12.3, not new derivation. v1.1 PARTIAL status preserved.
NC-Q9.9: Functor F in Q9.6 is universality-class equivalence, not strict.
NC-Q9.10: v1.2 is mathematical strengthening, not experimental adjudication. Data analysis is v1.3+ content.
NC-Q9.11: v1.1 MC margin (~76×) reported honestly; v1.0 ~700× projection was independence-assumption overestimate.
NC-Q9.12 (v1.2 new): Lemma Q9.3a (Oriented Conjugate Derivative) is a CORPUS-INHERITED structural lemma, derived from ZS-S6 §4.1 + ZS-M32 §1.3 K_bwd ≠ K_fwd^† PROVEN structure, not a new free assumption.
NC-Q9.13 (v1.2 new): PRN Convention B (f-domain) fixing is a CONVENTION CHOICE for unambiguity, not a physical claim. Convention A (ω-domain) is equally valid; results are related by $(2\pi)^2$ rescaling (Appendix C).
NC-Q9.14 (v1.2 new): Theorem Q9.7 establishes diffeomorphism status of the reconstruction map; this is the MATHEMATICAL STATUS, not a claim of physical injectivity (which depends on operating-point selection).

§16. Conclusion and Outlook

ZS-Q9 v1.2 is the mathematically strengthened framework that elevates the i-tetration fixed point z* from a corpus-internal algebraic identity to a lab-measurable object via complex transmission time delay under PRN Convention B. The principal v1.2 advance over v1.1 is the closure of two mathematical weaknesses identified by feedback: (1) Theorem Q9.2 is redefined as the Two-Coordinate Spectral Reconstruction Bridge with explicit closed-form map C(R_lab, Φ_lab) = (2/π)·exp(-R_lab/2)·exp(i·Φ_lab); (2) Theorem Q9.3 is corrected via Lemmas Q9.3a/b that lift the corpus-PROVEN K_bwd ≠ K_fwd^† kernel structure to amplitude-level derivatives. Both fixes promote the respective theorems from DERIVED-CONDITIONAL (v1.1) to DERIVED (v1.2).

Three new theorems and one corollary extend the framework: Theorem Q9.7 (Reconstruction Map Smoothness) confirms diffeomorphism status; Theorem Q9.8 (Error Propagation) provides closed-form lab precision requirements; Theorem Q9.9 (Wilson Loop Specificity) converts the v1.1 Q9.5 PARTIAL FAIL into positive evidence that 0.7948 is measure-zero specific to Z-Spin spectra (100% of generic 11-pole spectra fail the target). Corollary Q9.4a (Z_10 Pentagonal Doubling Group) identifies $\alpha_{op} = \pi/5$ as the generator of Z_10, the group-theoretic doubling of the corpus-PROVEN Z_5 pentagon-tetration group.

PRN Convention B (f-domain) is fixed throughout, eliminating the v1.1 ω/f 2π factor ambiguity. The likelihood ratio reporting ($\ln \Lambda = 8.94$, decisive evidence on Jeffreys 1939 scale) supplements the v1.1 p-value reporting ($p_{LEE} = 1.31 \times 10^{-4}$, ~76× margin).

Three timeline milestones are pre-registered:
(M1) v1.3 (target Q3 2026): Giovannelli S(f) real data extraction with bootstrap ρ(L1, L4) analysis → single-dataset HYPOTHESIS-strong closure target. Theorem Q9.9 power: < 1/1000 generic match probability.
(M2) v1.4 (target Q4 2026): Asano raw data joint analysis with both datasets → HYPOTHESIS-strong promoted closure target.
(M3) v2.0 (target 2027): All L1-L6 closure on both datasets → DERIVED-CONDITIONAL closure.
(M4) v2.1 (long-term): Third-system confirmation (atomic photoionization or photonic crystal) → DERIVED-strong promotion.

The v1.2 falsification gates F-Q9.1 to F-Q9.7 (BLOCKING) and F-Q9.6 (CONFIRMING) provide explicit experimental falsifiability paths. The honest preservation of Q9.5 PARTIAL status while extracting positive evidence via Theorem Q9.9 exemplifies the Z-Spin epistemic standard: negative intermediate results are structural information, not failures.

Acknowledgements
The author thanks the Z-Spin Collaboration internal review team for the v1.0, v1.1, and v1.2 pre-publication audits. The v1.1 → v1.2 mathematical strengthening (Two-Coordinate Reconstruction, Lemmas Q9.3a/b, PRN Convention B fixing, three new theorems) was made possible by the comprehensive external mathematical feedback received on v1.1 in May 2026, which correctly identified the sign error in v1.1 Q9.3 and the trace-log insufficiency in v1.1 Q9.2. All feedback recommendations were adopted in v1.2. v1.2 numerical verification was performed with mpmath at 50-digit precision (Tests C, D, D2, E, F1, F2, G) and at 30-digit Python random for the v1.1 1.5×10^6-trial MC. External data is gratefully acknowledged from M. Asano et al. (Nat. Commun. 7, 13488) and from I. L. Giovannelli and S. M. Anlage (Phys. Rev. Lett. 135, 043801).

Code Availability
Pre-registration code (zs_q9_v1_precheck.py, zs_q9_v11_tests.py, zs_q9_v12_precheck.py, zs_q9_verify_v1_1.py — all at 50-digit mpmath), analysis pipeline templates for both datasets, and JSON verification results are committed to the public Z-Spin Collaboration repository at https://github.com/KennyKang-git/zspin/tree/main/papers/06_Quantum_Mechanics/ZS-Q9. All gates, baskets, tolerances, and random seed (20260517) are frozen at the v1.2 commit hash. The v1.2 verification suite extends v1.1 by 7 new tests (C2, D2, E11, F1, F2 + 2 functor diffeomorphism checks).

Appendix A. Verification Suite (v1.2 execution)

52 verification tests in 8 categories (v1.2 extends v1.1 by 7 new tests). 49 PASS; 3 PARTIAL (Cat. F.Q9.5 synthetic, honestly downgraded since v1.1).

Table A.1. v1.2 verification suite by category.

Cat.	Test description	Result	Max residual or note
A. L1-L5	Self-locking identities of ZS-M1 §3	5/5 PASS	5.68×10^{-27}
B. z*	i-tetration convergence to z*	6/6 PASS	$< 10^{-40}$
C. Q9.2	Two-Coordinate Reconstruction Bridge	7/7 PASS	C2 residual $< 5 \times 10^{-32}$
D. Q9.3	Im(V_ZY · V_XZ) = 0 + Q9.3 v1.2 corrected	7/7 PASS	D2 sum residual 2.98×10^{-26}
E. Q9.4	k=1..10 phase-doubling + Z_10 closure	11/11 PASS	E11 closure exactly 0
F. Q9.5/Q9.7/Q9.8	Synthetic SFF + new theorems Q9.7, Q9.8	3/3 PASS (Q9.7,Q9.8); 0/5 PARTIAL (Q9.5)	F1 |det J| match exact; F2 σ_z match exact
G. Q9.6/Q9.9	F1/F2/F3/F4 functor + Q9.9 Specificity	11/11 PASS	Q9.9: 0/1000 generic match
H. MC	3-basket × 500K anti-numerology	STRONG PASS	p_LEE = 1.31×10^{-4}; ln Λ = 8.94

A.1 Cat. A: L1-L5 Residuals (preserved from v1.1)

L1: 3.389×10^{-27}. L2: 1.583×10^{-27}. L3: 3.057×10^{-27}. L4: 5.683×10^{-27}. L5: strict inequality, margin 0.0691.

A.2 Cat. C: Theorem Q9.2 v1.2 Two-Coordinate Reconstruction

C1: tr_Z[ln M_f] = 2 ln|λ| = -0.22966924999... at 50-digit (preserved from v1.1).
C2 (v1.2 new): Reconstruction C(R_lab, Φ_lab) at 50-digit yields z* with residual |C - z*| $< 5 \times 10^{-32}$.
R_lab = 0.229669... and Φ_lab = 0.688453... rad = 39.4455°.
C3 (v1.2 new): Inverse map $C^{-1}(z^*)$ recovers (R_lab, Φ_lab) with residuals $< 10^{-51}$.
C4 (v1.2 new): Diffeomorphism check at 5 random (R, Φ) points: all round-trip residuals $< 5 \times 10^{-51}$.

A.3 Cat. D: Theorem Q9.3 v1.2 with Lemmas Q9.3a/b

D1: Im(V_ZY · V_XZ) = 0 exactly at 100 lattice points (preserved from v1.1).
D2 (v1.2 new): Im[∂_f^fwd ln V_XZ] + Im[∂_f^bwd ln V_ZY] = -π exactly, residual 2.98×10^{-26}.
D3 (v1.2 new): Δρ|_op = -1/2 (signed), |Δρ|_op| = 1/2, residual 5×10^{-26}.

A.4 Cat. E: Theorem Q9.4 v1.2 Phase-Doubling Cycle

E1-E10: preserved from v1.1, k=1..10 predictions at 50-digit precision.
E11 (v1.2 new, Corollary Q9.4a): 10·α_op = 6.28318530717958... and 2π = 6.28318530717958..., residual exactly 0. Z_10 group closure confirmed.

A.5 Cat. F: Q9.5/Q9.7/Q9.8

F1 (Q9.7 diffeomorphism, v1.2 new): |det J| at (R_lab, Φ_lab) = 0.16106..., residual from $-(2/\pi^2)\cdot\exp(-R)$ is exactly 0.
F2 (Q9.8 error propagation, v1.2 new): example calculation ($\sigma_R = 0.1$, $\sigma_\Phi = 2°$) → $\sigma_z = 0.0346$ in z-plane, residual from analytical formula 0.0.
F3-F7 (Q9.5 synthetic SFF): 0/5 PARTIAL (preserved from v1.1, residual 0.77).

A.6 Cat. G: Q9.6/Q9.9

G1-G10: functor F properties F1/F2/F3/F4 (preserved from v1.1, 10/10 PASS).
G11 (Q9.9 Specificity, v1.2 new): 0 of 1000 random 11-pole spectra fall within ±0.05 of 0.7948. 100% generic FAIL rate. Confirms Z-Spin Wilson loop specificity.

A.7 Cat. H: Anti-Numerology MC (preserved from v1.1)

Seed = 20260517. Basket H1: 0.0526%. Basket H2: 0.0234%. Basket H3: 0.0014%.
p_LEE = 1.31×10^{-4} (76× margin over Z-Spin 1% threshold).
v1.2 new likelihood ratio: $\ln \Lambda = 8.94$ (decisive evidence per Jeffreys 1939 scale).

Appendix B. Data Provenance Audit

Table B.1. External data sources and provenance.

Source	Type	Use in ZS-Q9
Asano et al. 2016 [3]	Optical, 17-ns pulse, $Q_0 \approx 2.9\times10^6$	Dataset A: Q9.1, Q9.4, Q9.6
Giovannelli-Anlage 2025 [4]	Microwave, 5.27 GHz, γ_3dB=11.15 MHz	Dataset G: Q9.1, Q9.4, Q9.6
Smith 1960 [1]	Theory: Wigner-Smith Q matrix	External reference (§2.1)
Guo-Gasparian 2022 [2]	Theory: Krein-Friedel sub-unitary	Foundation for Q9.3 (§6)
Patel-Michielssen 2021 [5]	Theory: WS for electromagnetism	Foundation for Q9.5 (§8)
AAV 1988 [6]	Theory: weak value	Foundation for ι (Q8 inheritance)

B.1 PRN Integrity Audit Trail (preserved from v1.1)

γ_3dB = 11.15 MHz for Giovannelli is published in [4] Fig. 2(a) inset; PRN integrity satisfied by public availability. γ_optical for Asano requires independent extraction from raw data (v1.3 target).

Appendix C. ω vs f Convention Cross-Reference (v1.2 new)

v1.2 fixes Convention B (f-domain) throughout the body. For cross-checking against external literature that uses ω-domain (rad/s), Appendix C documents the conversion table.

Table C.1. Convention B (f) ↔ Convention A (ω) conversion.

Quantity	Convention B (f, used)	Convention A (ω, NOT used)
Linewidth γ_3dB	11.15 MHz (in Hz)	Γ_ω = 2π × 11.15 MHz = 70.06 Mrad/s
Pulse FWHM Δ̃f	5 MHz (in Hz)	Δ̃ω = 2π × 5 MHz = 31.4 Mrad/s
Frequency shift D_f	0.48 MHz (in Hz)	D_ω = 2π × 0.48 MHz = 3.0 Mrad/s

Quantity	Convention B (f, used)	Convention A (ω, NOT used)
Time delay τ	$\tau_f = -(i/2\pi)\cdot\partial_f \ln T$ (seconds)	$\tau_\omega = -i\cdot\partial_\omega \ln T$ (seconds, same)
PRN dimensionless Z_{exp}	$\gamma_{3dB} \cdot \tau_f$	$\Gamma_\omega \cdot \tau_\omega = (2\pi\cdot\gamma_{3dB}) \cdot \tau_\omega = (2\pi)^2 \cdot (\gamma_{3dB} \cdot \tau_f)$
Conversion factor	1 (reference)	$(2\pi)^2 \approx 39.48$

Critical note. Using Convention A consistently and Convention B consistently are both mathematically valid. The DANGER is mixing the two — e.g., using ω-domain τ definition with f-domain numerical γ. v1.1 contained such a latent ambiguity in its PRN equation. v1.2 eliminates the risk by explicit Convention B fixing.

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Version History

v1.0 (March 2026): Initial public release. Theorem Q9.1 DERIVED-CONDITIONAL. Theorems Q9.2-Q9.6 introduced. Six-locking-gate hierarchy and PRN integrity protocol pre-registered. Verification: 36/36 PASS projection. Anti-numerology MC: protocol pre-registered, execution deferred.

v1.1 (May 2026): Complete textual derivation of all six theorems. Actual execution of anti-numerology MC (1.5×10^6 trials) with honest p_LEE = 1.31×10^{-4} reporting (~76× margin, revised from v1.0 projected ~700×). Q9.2 verified at 50-digit (residual 3.3×10^{-51}). Q9.3 verified at 100 lattice points (Im(V_ZY · V_XZ) = 0.0 exactly). Q9.4 k=1 to k=10 cycle closure. Q9.5 synthetic 11-pole test FAILED ±0.05 tolerance; status downgraded to PARTIAL (honest reporting). Q9.6 functor F1/F2/F3/F4 all 10/10 PASS. 44/45 PASS overall.

v1.2 (May 2026, this version): Mathematical strengthening incorporating all 14 items of v1.1 feedback. Theorem Q9.2 redefined as Two-Coordinate Spectral Reconstruction Bridge with explicit closed-form map C(R_lab, Φ_lab); status promoted from DERIVED-CONDITIONAL to DERIVED. Theorem Q9.3 corrected via Lemmas Q9.3a (Oriented Conjugate Derivative) and Q9.3b (K_bwd Corpus Inheritance); status promoted from DERIVED-CONDITIONAL to DERIVED. PRN Convention B (f-domain) explicitly fixed throughout; Convention A (ω-domain) documented in Appendix C for cross-reference. Title cover corrected: "Paper 9 of 8" → "Paper 9 | Quantum Mechanics Extension Series". Three new theorems: Q9.7 (Reconstruction Map Smoothness, DERIVED), Q9.8 (Error Propagation, DERIVED), Q9.9 (Wilson Loop Specificity, DERIVED — 100% generic FAIL rate). One new corollary: Q9.4a (Z_10 Pentagonal Doubling Group, DERIVED). Likelihood ratio reporting added (ln Λ = 8.94, decisive evidence per Jeffreys 1939). L1, L4 bootstrap correlation protocol P5 registered for v1.3. New NC-Q9.12, NC-Q9.13, NC-Q9.14. 52 verification tests, 49 PASS + 3 PARTIAL (Q9.5 synthetic, preserved from v1.1).

v1.3 (target Q3 2026): Real Giovannelli S(f) data extraction for L6/Q9.5 closure; ρ(L1, L4) bootstrap analysis; single-dataset HYPOTHESIS-strong closure target.
v1.4 (target Q4 2026): Dual-dataset analysis with Asano raw data → HYPOTHESIS-strong promoted closure target.
v2.0 (target 2027): All L1-L6 closure on both datasets → DERIVED-CONDITIONAL closure target.
v2.1 (long-term): Third-system confirmation → DERIVED-strong promotion.